

FREUID Challenge 2026 — Technical Report

1. Introduction

We address fraud detection in identity documents under a cross-domain private test regime (two undisclosed document types + recaptured images). Our guiding finding: **fine-tuned backbones overfit the seen countries and collapse on blind domains, while frozen pretrained backbones with lightweight LoRA adapters preserve the pretraining generalization prior.** The final system is a rank-mean ensemble of three frozen-backbone classifiers with an inference-side lever for recaptured images.

2. Method

2.1 Models

- **FB** — frozen DINOv2-L (`vit_large_patch14_dinov2.lvd142m`), LoRA on q/k/v of the last 8 blocks ($r=16$, $\alpha=32$), MLP head (hidden 256), input 448x714 (aspect resize), 5 folds. LoRA merged into plain weights after training.
- **FC** — frozen OpenCLIP ViT-L/336 (`vit_large_patch14_clip_336.openai_ft_in12k_in1k`), identical recipe, CLIP normalization. Decorrelated pretraining prior (image-text contrastive vs. self-supervised) is the source of the ensemble gain.
- **FD** — frozen SigLIP2-L/16-384 (`vit_large_patch16_siglip_384.v2_webli`), identical recipe, input 448x720, 0.5/0.5 normalization. A third, independently-pretrained prior (sigmoid image-text, WebLI).

All backbones are trained as 5-fold CV. The final pick uses all **5 FB folds** with hflip-TTA as its core; **FC3** (3 folds) and **FD3** (3 folds) score the captured subset only, so the captured ens3 reorder stays within the 6h/A100 inference cap (Sec. 4).

Blind leave-country-out (LTO) evidence for the frozen-vs-finetuned choice: fine-tuned FA collapses to 0.18 on held-out GUINEA; frozen FB reaches 0.030. Adding FC improves blind rank-mean by -35% (GUINEA) / -27% (EGYPT) and pooled OOF by -70%. Adding FD improves blind rank-mean further (-4.6% GUINEA / -54% EGYPT vs. the FB+FC pair; FD is the strongest single backbone on held-out EGYPT) and pooled OOF by another -57%.

2.2 Training

- Data: official FREUID training set ONLY (no external datasets for any submitted weight; pretrained backbones are Apache-2.0, see Sec. 7).
- 5-fold CV. Loss: BCE. AMP. Epoch budget 4 with early best-selection (frozen-backbone LoRA converges at epoch 0-1).
- Augmentation: standard photometric/geometric domain aug + tamper-synth ($p=0.3$) + G1 recapture-style aug. Heavier domain-randomization

(FACT amplitude-spectrum mix, Shades-of-Gray color constancy)
DESTROYED the fraud signal in ablation (train loss stuck at random plateau) and was rejected.

2.3 Inference pipeline

1. Per-backbone fold plain mean -> per-backbone score.
2. **Cross-backbone equal rank-mean over FB/FC/FD** (score-mean dilutes the weaker-calibrated backbone on out-of-domain data).
3. **Recaptured-row detection** by resolution: a resolution is native only if it is high-resolution (width ≥ 1000 px) AND either frequent ($\geq 0.5\%$ of rows) or a known native cluster; everything else — including frequent but low-resolution formats such as 840x530 — is captured. In the training data digital captures are always ≥ 1000 px wide while recaptured/downscaled images are smaller, so width cleanly separates the two acquisition types (a frequent 840x530 recapture format would otherwise be misread as native).
4. **capShift**: captured rows are shifted by $\Delta = 0.75$ in logit space (per backbone) — recaptured bona fide images otherwise inflate the fraud tail (APCER at 1% BPCER dominates the metric). Public-LB verified: 0.01870 -> 0.01524.
5. **Captured-internal reorder (ens3)**: within captured rows, order by the equal mean-rank of the three backbones (FB5 + FC3 + FD3). FC3/FD3 are scored on the captured subset only, in a TTA / noTTA / off tier chosen by the remaining inference budget at the observed captured fraction (captured ordering is robust to the TTA reduction, spearman 0.9958); capShift is CPU-side and always applies. The reorder mechanic is public-LB verified (0.01524 -> 0.01304 -> 0.01252 as the captured-reorder ensemble grew); the shipped ens3 orders captured rows by the three released backbones only (FB5 + FC3 + FD3). The intermediate 0.01304 / 0.01252 points additionally used recapture-specialist checkpoints that are not part of this release; the released package reproduces the ens3 composition, which keeps the captured lever within the inference budget.
6. Output = strict total order -> $(\text{pos}+0.5)/n$ rank values (metric is rank-only; ties score as constants).

The single final pick (**fb5**) runs this pipeline with an FB5 hflip-TTA core: step 1 aggregates the 5 FB folds, step 2 degenerates to the FB score order, and FC3/FD3 enter only through the captured reorder in step 5.

Precision: bf16 (rank-safety spearman vs fp32: FB 0.99992, FC 0.999799, FD 0.999627). Ensemble-level / cross-backbone TTA on the core was rejected by measurement — blind leave-country-out showed degradation (+27.8% on held-out EGYPT; hflip severely harms the CLIP backbone on documents). The FB5 core keeps hflip-TTA (benign for the DINOv2 backbone; part of the pick's frozen public composition), while FC3/FD3 touch only the captured subset.

3. Data

- Training: FREUID official training set (69,352 images), 5-fold splits. **No external dataset was used at any stage — not for training, and not for model selection.** The submitted weights and every selection

decision rely on the competition data (and public-leaderboard feedback) only.

- External model weights (pretrained backbones only): Sec. 7.
- Validation protocol: blind leave-country-out (GUINEA fold, EGYPT fold) — the only offline signal that correlated with cross-domain behavior; pooled OOF does NOT predict LB (measured: best-OOF model was worst on LB).

4. Inference / Budget

- Evaluator (organizer-confirmed): 1x NVIDIA A100 40 GB, 24 CPU cores, ≤ 6 h on the hidden test set.
- End-to-end offline (`--network none`) measurement on a single A10G (a ~ 2 - $2.7\times$ slower proxy for the A100): 7,821 images in ~ 33 min for the full pipeline (~ 0.25 s/img). Projected to the full 142,818-image test set: ~ 10 h on A10G -> **~ 5 h on A100 at a conservative /2.0 factor** (~ 3.5 h at a realistic /2.7), within the 6 h budget. Peak GPU memory < 23 GB (fits A100 40 GB).
- `torch.compile` is applied to the inference model (rank-preserving; raw scores shift $\sim 1e-2$ but the pipeline is rank-based end-to-end). It amortizes over the many batches of the full test set ($\sim +17\%$ throughput measured on A10G).
- The FC3/FD3 captured pass (25.8 / 27.7 ms/img/ckpt on A10G) touches captured rows only, in the budget tier of Sec. 2.3 step 5 (TTA up to $c \leq 9.8\%$, noTTA up to $c \leq 19.6\%$, off beyond — `capShift` still applies). Core cost (FB5 hflip-TTA) dominates and scales linearly with the image count, so the run stays within the cap regardless of the private captured fraction.

5. Results

Stage	Public LB
FB5 (frozen DINOv2-L + LoRA)	0.01870
+ <code>capShift</code> (delta 0.75)	0.01524
+ captured reorder, early ensemble ⁽¹⁾	0.01304
+ captured reorder, all backbones ⁽¹⁾	0.01252
+ captured ens3 reorder (released FB5+FC3+FD3, size-based routing)	0.01238
Container-native output (<code>docker run</code> , rank values, reproduces the above)	0.01245
Private fb5 pick (FB5 hflip-TTA + captured lever)	hidden until 2026-07-24

⁽¹⁾ These two intermediate public-LB points additionally used recapture-specialist checkpoints that are not part of this release. They are shown only to document the reorder mechanic's progression; the released package reproduces the ens3 (FB5+FC3+FD3) composition.

Blind LTO (held-out country, freuid score, lower=better):

Model	GUINEA	EGYPT
FA (fine-tuned)	0.18	—
FB (frozen)	0.0297	0.0647
FC (frozen)	0.0325	0.0656
FD (frozen)	0.0813	0.0143
FB+FC rank-mean	0.0194	0.0474
FB+FC+FD rank-mean	0.0185	0.0218

FD's country profile is the mirror of FB/FC (weakest on GUINEA, 4.5x strongest on EGYPT) — direct evidence of prior decorrelation; the 3-way rank-mean improves both held-out countries.

The public leaderboard rows above keep the FB-based ordering: the public main block is in-domain, where cross-backbone rank-mean measurably loses to the single-backbone/score-mean orderings (consistent with pooled OOF); the 3-backbone rank-mean targets the cross-domain private blocks, per the blind LTO table.

6. Reproducibility

- Repo: <https://github.com/hyun910219/freuid-challenge-2026>, commit SHA: 813c68f3a30cb1251b51bddce1fc195ea4fbcebe. Model weights are frozen throughout; every post-freeze commit is inference-orchestration / documentation only (size-based captured routing; the `--shm-size` runtime requirement + a `num_workers=0` fallback; the `VARIANT` flag selecting the captured-reorder backbones). No weight/architecture/training change at any point.
- Runtime: run with `--shm-size=16g` (or `--ipc=host`). DataLoader worker->main tensor IPC uses `/dev/shm` even with the `file_system` sharing strategy on torch 2.12; the 64 MB docker default is exhausted otherwise. The entrypoint falls back to `num_workers=0` when the flag is absent (correct but slower).
- Two final picks from ONE image via a documented inference-time flag (host-approved: same commit + weights, flag only). Submission <-> command <-> output checksum:

Final pick (Kaggle id / timestamp)	Command (all <code>--network none --shm-size=16g</code>)	sha256(container submission.csv)
Pick 1 — submission_fb5_fc3fd3_SUBMIT.csv / 2026-07-14 02:55 UTC	<code>docker run ... freuid- repro:local (VARIANT=ens3 , captured = FB5+FC3+FD3)</code>	<code>c09feef...908fc77e</code>
Pick 2 — submission_fb5_fd3_SUBMIT.csv / 2026-07-14 03:14 UTC	<code>docker run ... -e VARIANT=fd freuid-</code>	<code>ac7eb76...eab33efe</code>

Final pick (Kaggle id / timestamp)	Command (all --network none --shm- size=16g)	sha256(container submission.csv)
	repro:local (captured = FB5+FD3)	

(Reference sha256 from our A10G run; a fresh run matches at rank-Spearman

= 0.9999 — bf16/compile perturbs raw scores $\sim 1e-2$, the pipeline is rank-based.)

Per the organizers, only the ranking submission (the better of the two selected picks) must reproduce; both are documented. The container output's public block and row order differ from the uploaded CSV (public rows were frozen mid- competition, see the note on the submitted file); the private-block ranking — which decides the score — is what the container reproduces. -

Docker: no-network container, weights baked in; flat /data in, /submissions/ submission.csv out. See README. Verified end-to-end (git clone -> docker build -> docker run --network none) on public images -> valid, contract-compliant submission.csv offline. - Training: one config per member, src/train.py --config configs/<m>.yaml. - Deterministic postprocessing; GPU/compile nondeterminism shifts raw scores at $\sim 1e-2$ but the pipeline is rank-based (the offline container reproduces our submitted ranking at Spearman ≥ 0.9999).

7. External Resources

Resource	License	Use
DINOv2 ViT-L 1vd142m (timm/Meta)	Apache-2.0 (repo LICENSE verified)	FB backbone
OpenCLIP ViT-L/336 openai_ft_in12k_in1k (timm)	Apache-2.0 (OpenAI CLIP origin: MIT)	FC backbone
SigLIP2 ViT-L/16-384 v2_webli (timm/big_vision)	Apache-2.0 (big_vision repo LICENSE verified)	FD backbone
PyTorch 2.12 / timm 1.0.27 / albumentations 2.0.8 etc.	BSD/Apache/MIT	framework

Only the pretrained backbones above (all license-compatible) and the listed frameworks are used. No external dataset was used at any stage — not to train any submitted weight, and not for model selection.